



Name: _____ Student No.: _____ Course/Section: _____ Date: _____



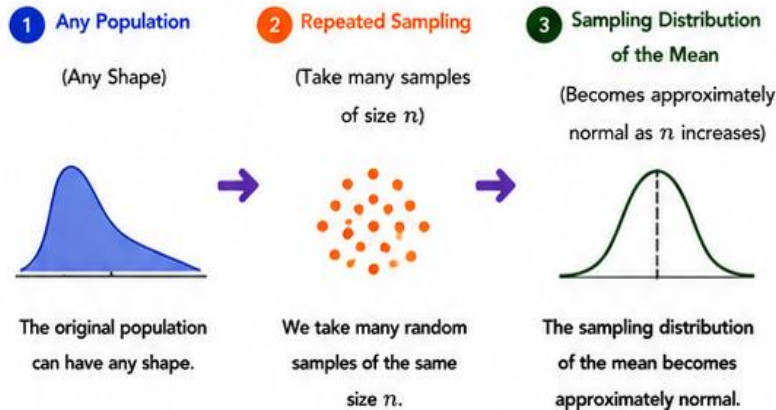
A. CONCEPT CHECK

Read each part carefully and answer the questions.



The Central Limit Theorem (CLT) states that as the sample size increases, the sampling distribution of the sample mean becomes approximately normal, regardless of the shape of the original population distribution.

How CLT Works



In simple words:

Even if the original population is not normal, large enough samples will produce sample means that follow a normal distribution.

1 FILL IN THE BLANKS. Complete the statements using the correct words.

sample size approximately normal any shape sample mean increases

- The Central Limit Theorem applies to populations of _____.
- As the _____ increases, the sampling distribution of the _____ becomes _____.
- The sampling distribution described by CLT is the distribution of the _____.
- This happens because we take many samples and compute their means, and as n _____, those means follow a normal distribution.

2 IDENTIFY THE CONCEPT. Write whether the statement describes LLN, CLT, or Both.

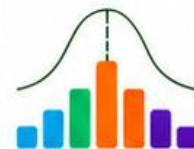
LLN = Law of Large Numbers
CLT = Central Limit Theorem

- Describes how the sample mean gets closer to the population mean as sample size increases. _____
- States that the sampling distribution of the mean becomes approximately normal as sample size increases. _____
- Applies regardless of the shape of the population distribution. _____
- Focuses on convergence of the sample mean to the true mean. _____

3 SHORT ANSWER QUESTIONS.

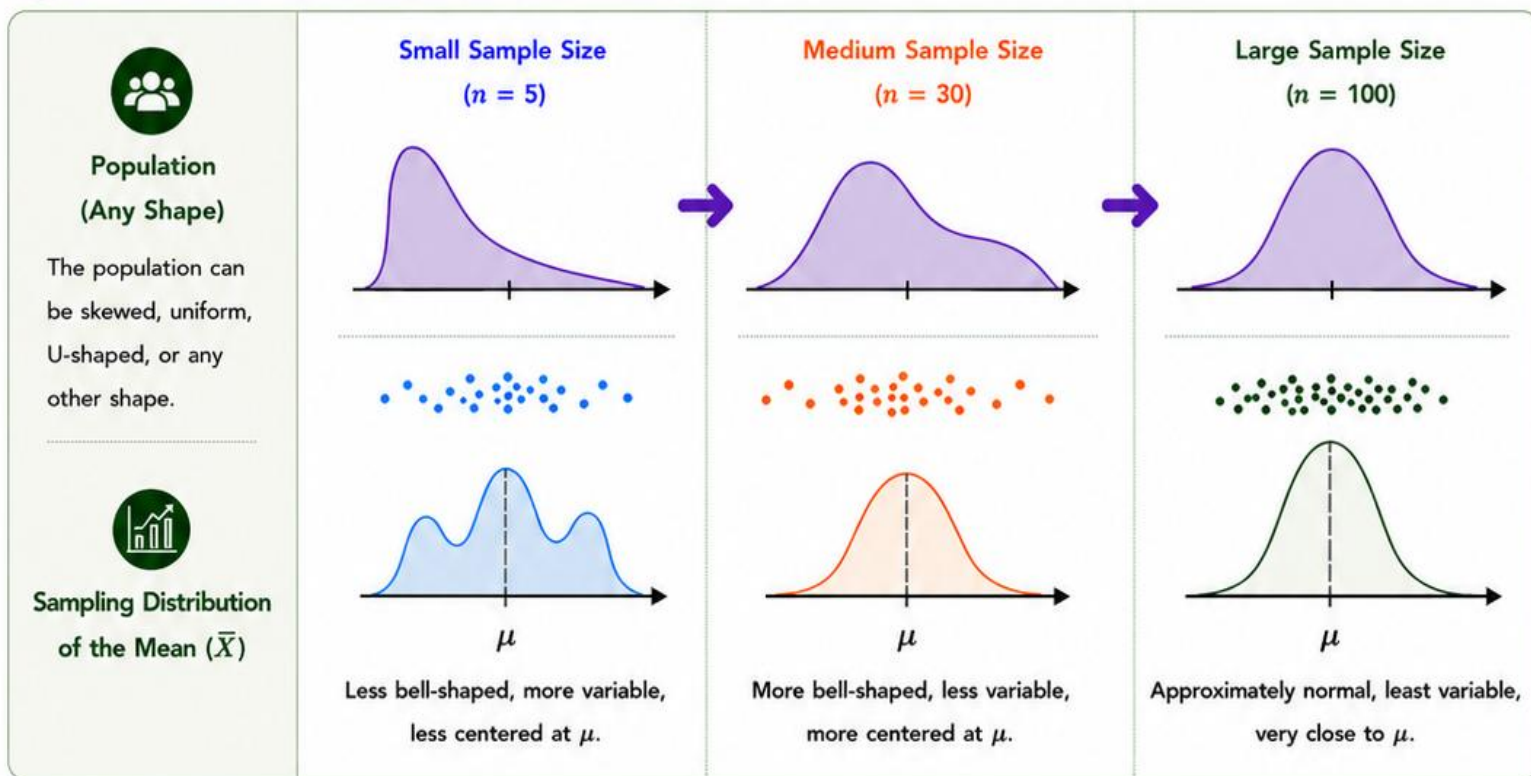
- Why can the original population have any shape for the Central Limit Theorem to still work?

- What do we mean by "sampling distribution of the mean"?



A. FROM ANY SHAPE TO APPROXIMATELY NORMAL

Study the diagrams below and answer the questions.



1 INTERPRET THE DIAGRAMS. Complete the table below.

Aspect	Small Sample Size ($n = 5$)	Medium Sample Size ($n = 30$)	Large Sample Size ($n = 100$)
Shape of sampling distribution of $\bar{X}$		more bell-shaped	
Variability (spread) of sample means	more variable		
Closeness of sample mean to μ		more centered at μ	
Overall description			approximately normal and very consistent

2 ANSWER THE QUESTIONS.

- How does the shape of the sampling distribution of the mean change as the sample size increases?

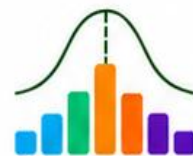
- How does the variability (spread) of sample means change as the sample size increases?

- Why do sample means become closer to μ as the sample size increases?

3 CONNECT THE IDEAS. Fill in the blanks using the words in the box.

larger | approximately normal | less variable | centered

As the sample size (n) becomes _____, the sampling distribution of the mean becomes _____, it is _____, and it is more _____ around μ .



A. COMPARE LLN AND CLT *Study the table and answer the questions.*

Aspect	Law of Large Numbers (LLN) 	Central Limit Theorem (CLT) 
What it states	As sample size increases, the sample mean gets closer to the population mean (μ).	As sample size increases, the sampling distribution of the mean becomes approximately normal.
Focus	Convergence of $\bar{X}$ to μ	Shape (normality) of sampling distribution
Depends on the shape of population	No	No (works regardless of shape)
Main result	$\bar{X}$ converges to μ	Distribution of $\bar{X}$ becomes normal
When it is about	Relates to the value of the mean	Relates to the distribution of the mean

HELPFUL TIP!

LLN is about where the mean is going.

CLT is about what the mean looks like.

1 IDENTIFY THE CONCEPT. Write LLN, CLT, or Both.

- Describes how the sample mean gets closer to the population mean as sample size increases. _____
- States that the sampling distribution of the mean becomes approximately normal. _____
- Applies regardless of the shape of the original population. _____
- Focuses on the shape of the sampling distribution, not just the value of the mean. _____
- Explains why $\bar{X}$ converges to μ . _____

2 SCENARIO CHECK. Decide which concept (LLN, CLT, or Both) best describes each situation.

Situation	LLN / CLT / Both
1 When we take many samples, the average of the averages gets closer to the true mean.	
2 The sampling distribution of the mean becomes bell-shaped as n increases.	
3 Works even if the population is skewed.	
4 Explains why $\bar{X}$ approaches μ as n gets larger.	
5 Tells us the shape of the sampling distribution of the mean.	

QUICK VISUAL CHECK!

Which concept does each statement describe? (Write LLN, CLT, or Both)

- The mean of the sample converges to μ .

- The sampling distribution becomes normal.

- Works regardless of the population shape.

3 TRUE OR FALSE. Write T if the statement is true and F if it is false.

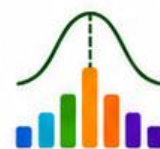
- LLN describes the shape of the sampling distribution. _____
- CLT explains why the sampling distribution becomes approximately normal when n is large. _____
- Both LLN and CLT work regardless of the population shape. _____
- LLN and CLT are the same concept. _____
- LLN is about the convergence of the sample mean to the true mean. _____

DID YOU KNOW?

LLN tells us
"where it goes."

CLT tells us
"what it looks like."

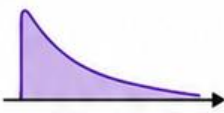
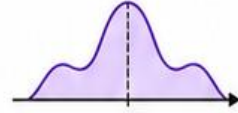
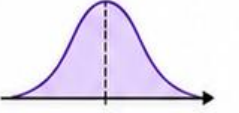
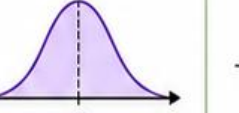
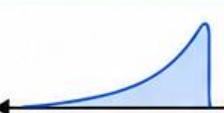
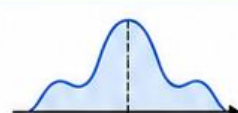
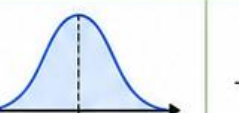
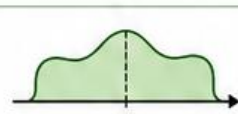
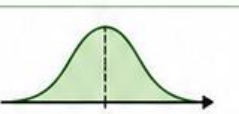
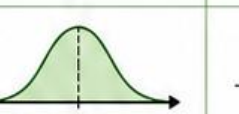




PRACTICE AND APPLY

Solve or answer the following.

- 1 SHAPE TRANSFORMATION.** For each original population shape, describe how the sampling distribution of the mean changes as n increases from 5 to 30 to 100.

Original Population	$n = 5$ (Small)	$n = 30$ (Medium)	$n = 100$ (Large)	What happens as n increases?
a)  Right-skewed				_____
b)  Left-skewed				_____
c)  Uniform				_____

- 2 EFFECT OF SAMPLE SIZE.** Choose the best answer.

- Which statement best describes what happens to the variability of the sampling distribution of the mean as n increases?
 - It increases.
 - It decreases.
 - It stays the same.
 - It becomes zero.
- Which of the following is true as the sample size becomes large?
 - The sampling distribution becomes more skewed.
 - The sampling distribution becomes approximately normal.
 - The sampling distribution becomes more spread out.
 - The sampling distribution no longer depends on n .
- If the original population is highly skewed and we take a large sample, what can we say about the sampling distribution of the mean?
 - It will still be highly skewed.
 - It will be approximately normal.
 - It will be uniform.
 - It will have the same shape as the population.
- Which sample size is most likely to produce a sampling distribution that is approximately normal?

A. $n = 3$	C. $n = 30$
B. $n = 10$	D. $n = 100$

- 3 APPLICATION.** Determine whether the Central Limit Theorem (CLT) can be applied to each situation.

Write **Yes** or **No** and give a reason.

Situation	Can CLT be applied?	Reason
a) A box contains 50,000 screws. We take a sample of 80 screws to find the average length.	_____	_____
b) A small town has only 15 households. We survey 10 households about their monthly electricity usage.	_____	_____
c) A factory produces widgets. The weight distribution is unknown. A sample of 35 widgets is taken.	_____	_____
d) A teacher wants to estimate the average score of students in a class of 25 by sampling 20 students.	_____	_____



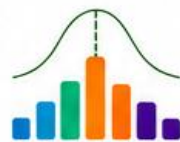
Reflection

What have you learned about the Central Limit Theorem? Write one key idea in your own words.



SELF-CHECK

- I can explain the CLT. ☐
- I can compare LLN and CLT. ☐
- I can apply CLT to situations. ☐
- I can describe how sample size affects the sampling distribution. ☐



1 THINK CRITICALLY. Answer the questions in complete sentences.

- The population of exam scores is highly skewed. We take samples of size $n = 50$. What can we say about the sampling distribution of the mean?

- If the sample size is increased from $n = 20$ to $n = 80$, how will the sampling distribution of the mean change?

- Can the Central Limit Theorem be applied if the sample size is very small, like $n = 3$? Explain your answer.

KEY REMINDER



- Larger $n \rightarrow$ more normal
- Larger $n \rightarrow$ less spread (less variable)
- Larger $n \rightarrow$ closer to μ

These three ideas describe what happens to the sampling distribution of the mean as n increases.

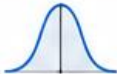


2 FILL IN THE BLANKS. Choose the correct word from the box to complete each statement.

normal population closer larger shape less variable Central Limit Theorem

- The _____ states that the sampling distribution of the mean becomes approximately normal as the sample size increases.
- This result holds regardless of the _____ (as long as the sample size is large enough).
- As the sample size becomes _____, the sampling distribution of the mean becomes more _____.
- It also becomes _____ to the population mean μ .
- The sampling distribution approaches a _____ distribution.

3 SUMMARY CHART. Complete the chart below.

Aspect	As sample size (n) increases, the sampling distribution of the mean...
 Shape	_____
 Variability (spread)	_____
 Closeness to μ	_____
 Overall description	_____

ONE LAST CHECK!

Before you finish, ask yourself:

- ☐ Can I explain what CLT is?
- ☐ Can I describe what happens as sample size increases?
- ☐ Can I apply CLT to situations?
- ☐ Do I understand the difference between LLN and CLT?